

Employee Attrition Analysis

Exploratory Data Analysis of Employee Attrition, Job Roles and Workplace Factors

1. Introduction

Employee attrition is an important business issue because employee turnover can affect workforce stability and recruitment requirements.

In this project, Exploratory Data Analysis (EDA) is performed on employee data to identify patterns related to attrition.

The analysis focuses on departments, job roles, overtime, job satisfaction, monthly income and years at the company.

2. Business Problem

Organizations need to understand employee attrition patterns in order to identify areas that may require attention.

This analysis compares employees who stayed and those who left using job roles, departments, overtime, job satisfaction, monthly income and years at the company.

The objective is to identify descriptive patterns in the dataset rather than predict individual employee attrition.

3. Objectives

- Understand the structure of the employee dataset
- Check missing values and duplicate records
- Perform basic data cleaning
- Analyze overall employee attrition
- Compare attrition across departments and job roles
- Analyze overtime and job satisfaction
- Compare monthly income and years at the company
- Examine the relationship between income and tenure
- Generate useful business insights

4. Import Libraries

```
In [98]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

5. Load Dataset

```
In [99]: df = pd.read_csv('/content/drive/MyDrive/Data Analytics/IBM_HR-Employee-Attrition.csv')
df.head()
```

```
Out[99]:
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	...
0	41	Yes	Travel_Rarely	1102	Sales	1	2	...
1	49	No	Travel_Frequently	279	Research & Development	8	1	...
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	...
3	33	No	Travel_Frequently	1392	Research & Development	3	4	...
4	27	No	Travel_Rarely	591	Research & Development	2	1	...

5 rows × 35 columns



6. Data Understanding

The dataset contains 1,470 employee records and 35 columns containing demographic, job-related and workplace information.

```
In [100]: print("Shape:", df.shape)

print("\nColumns:")
print(df.columns.tolist())

print("\nData Types:")
print(df.dtypes)

df.info()
```

Shape: (1470, 35)

Columns:

['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department', 'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount', 'EmployeeNumber', 'EnvironmentSatisfaction', 'Gender', 'HourlyRate', 'JobInvolvement', 'JobLevel', 'JobRole', 'JobSatisfaction', 'MaritalStatus', 'MonthlyIncome', 'MonthlyRate', 'NumCompaniesWorked', 'Over18', 'OverTime', 'PercentSalaryHike', 'PerformanceRating', 'RelationshipSatisfaction', 'StandardHours', 'StockOptionLevel', 'TotalWorkingYears', 'TrainingTimesLastYear', 'WorkLifeBalance', 'YearsAtCompany', 'YearsInCurrentRole', 'YearsSinceLastPromotion', 'YearsWithCurrManager']

Data Types:

```
Age                int64
Attrition          object
BusinessTravel     object
DailyRate          int64
Department         object
DistanceFromHome   int64
Education           int64
EducationField     object
EmployeeCount      int64
EmployeeNumber     int64
EnvironmentSatisfaction int64
Gender             object
HourlyRate         int64
JobInvolvement     int64
JobLevel           int64
JobRole            object
JobSatisfaction    int64
MaritalStatus      object
MonthlyIncome      int64
MonthlyRate        int64
NumCompaniesWorked int64
Over18             object
OverTime           object
PercentSalaryHike  int64
PerformanceRating  int64
RelationshipSatisfaction int64
StandardHours      int64
StockOptionLevel   int64
TotalWorkingYears  int64
TrainingTimesLastYear int64
WorkLifeBalance    int64
YearsAtCompany     int64
YearsInCurrentRole int64
YearsSinceLastPromotion int64
YearsWithCurrManager int64
```

dtype: object

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 1470 entries, 0 to 1469

Data columns (total 35 columns):

#	Column	Non-Null Count	Dtype
0	Age	1470 non-null	int64
1	Attrition	1470 non-null	object
2	BusinessTravel	1470 non-null	object
3	DailyRate	1470 non-null	int64
4	Department	1470 non-null	object
5	DistanceFromHome	1470 non-null	int64
6	Education	1470 non-null	int64
7	EducationField	1470 non-null	object
8	EmployeeCount	1470 non-null	int64

```

9   EmployeeNumber      1470 non-null   int64
10  EnvironmentSatisfaction  1470 non-null   int64
11  Gender              1470 non-null   object
12  HourlyRate          1470 non-null   int64
13  JobInvolvement      1470 non-null   int64
14  JobLevel            1470 non-null   int64
15  JobRole             1470 non-null   object
16  JobSatisfaction     1470 non-null   int64
17  MaritalStatus       1470 non-null   object
18  MonthlyIncome       1470 non-null   int64
19  MonthlyRate         1470 non-null   int64
20  NumCompaniesWorked  1470 non-null   int64
21  Over18              1470 non-null   object
22  OverTime            1470 non-null   object
23  PercentSalaryHike   1470 non-null   int64
24  PerformanceRating   1470 non-null   int64
25  RelationshipSatisfaction 1470 non-null   int64
26  StandardHours       1470 non-null   int64
27  StockOptionLevel    1470 non-null   int64
28  TotalWorkingYears   1470 non-null   int64
29  TrainingTimesLastYear 1470 non-null   int64
30  WorkLifeBalance     1470 non-null   int64
31  YearsAtCompany      1470 non-null   int64
32  YearsInCurrentRole   1470 non-null   int64
33  YearsSinceLastPromotion 1470 non-null   int64
34  YearsWithCurrManager 1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB

```

7. Missing Value Analysis

Missing values were checked across all columns. The dataset contains no missing values, so no imputation was required.

```

In [101... missing_values = df.isnull().sum()

print("Total Missing Values:", missing_values.sum())

```

Total Missing Values: 0

8. Duplicate Analysis

Duplicate records were checked to ensure that repeated employee records do not affect the analysis.

```

In [102... print("Duplicate Rows:", df.duplicated().sum())

```

Duplicate Rows: 0

9. Data Cleaning

Columns containing constant values or employee identification information were removed because they do not provide meaningful variation for the selected EDA.

```

In [103... df = df.drop(columns=[
    'EmployeeCount',
    'EmployeeNumber',
    'Over18',

```

```
'StandardHours'
])

df.shape
```

Out[103... (1470, 31)

10. Descriptive Statistics

Descriptive statistics were used to understand the numerical characteristics of employee income and company tenure.

Measures such as mean, median, minimum and maximum values were examined.

```
In [104... df[['MonthlyIncome',
      'YearsAtCompany']].describe()
```

Out[104...

	MonthlyIncome	YearsAtCompany
count	1470.000000	1470.000000
mean	6502.931293	7.008163
std	4707.956783	6.126525
min	1009.000000	0.000000
25%	2911.000000	3.000000
50%	4919.000000	5.000000
75%	8379.000000	9.000000
max	19999.000000	40.000000

11. Overall Employee Attrition

The overall distribution of employee attrition was analyzed to understand the proportion of employees who stayed with the organization and those who left.

This provides a baseline for comparing attrition across different employee groups.

```
In [105... attrition_count = df['Attrition'].value_counts()

print(attrition_count)

attrition_percentage = df['Attrition'].value_counts(normalize=True) * 100

print("\nAttrition Percentage:")
print(attrition_percentage.round(2))

attrition_count.plot(
    kind='bar',
    figsize=(7,5),
    title='Employee Attrition Distribution'
)

plt.xlabel('Attrition')
plt.ylabel('Number of Employees')
```

```
plt.xticks(rotation=0)
plt.show()
```

Attrition

No 1233

Yes 237

Name: count, dtype: int64

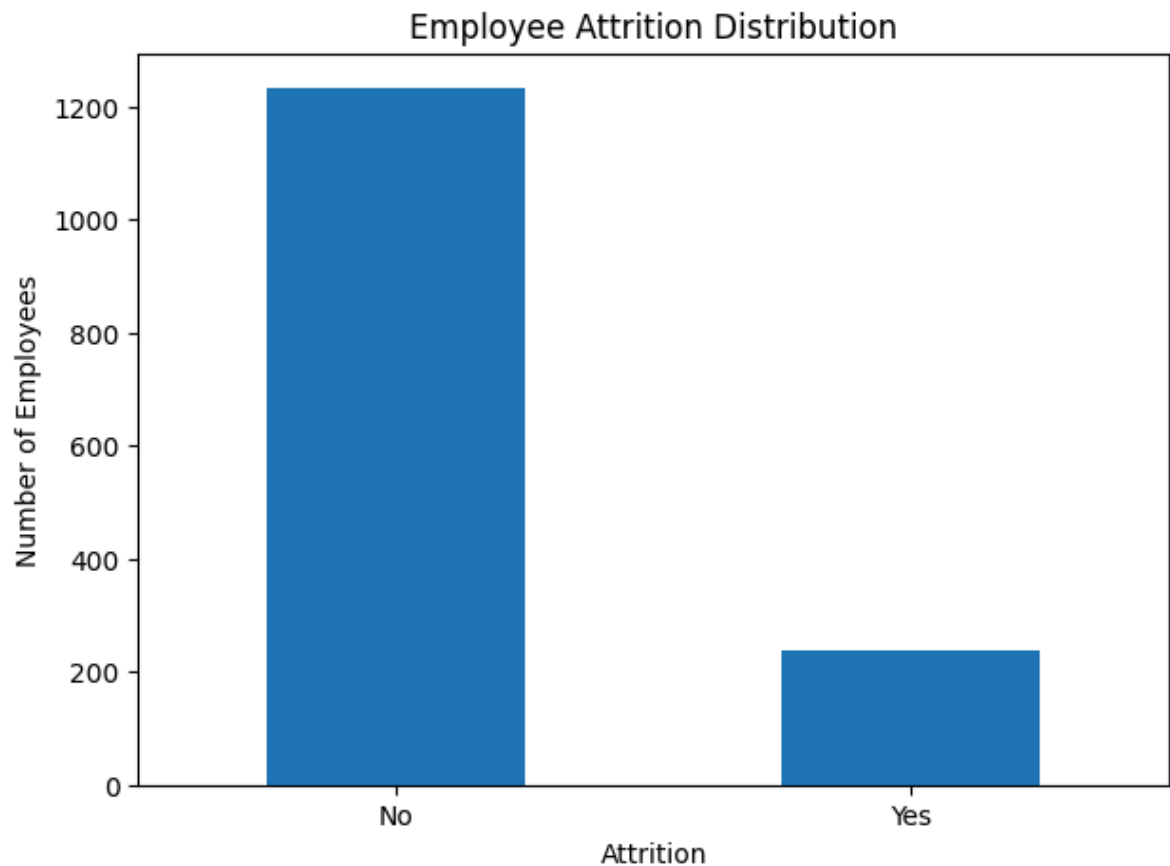
Attrition Percentage:

Attrition

No 83.88

Yes 16.12

Name: proportion, dtype: float64



12. Attrition by Department

Employee attrition was compared across departments using percentages to account for differences in department size.

```
In [106... department_attrition = pd.crosstab(
    df['Department'],
    df['Attrition'],
    normalize='index'
) * 100

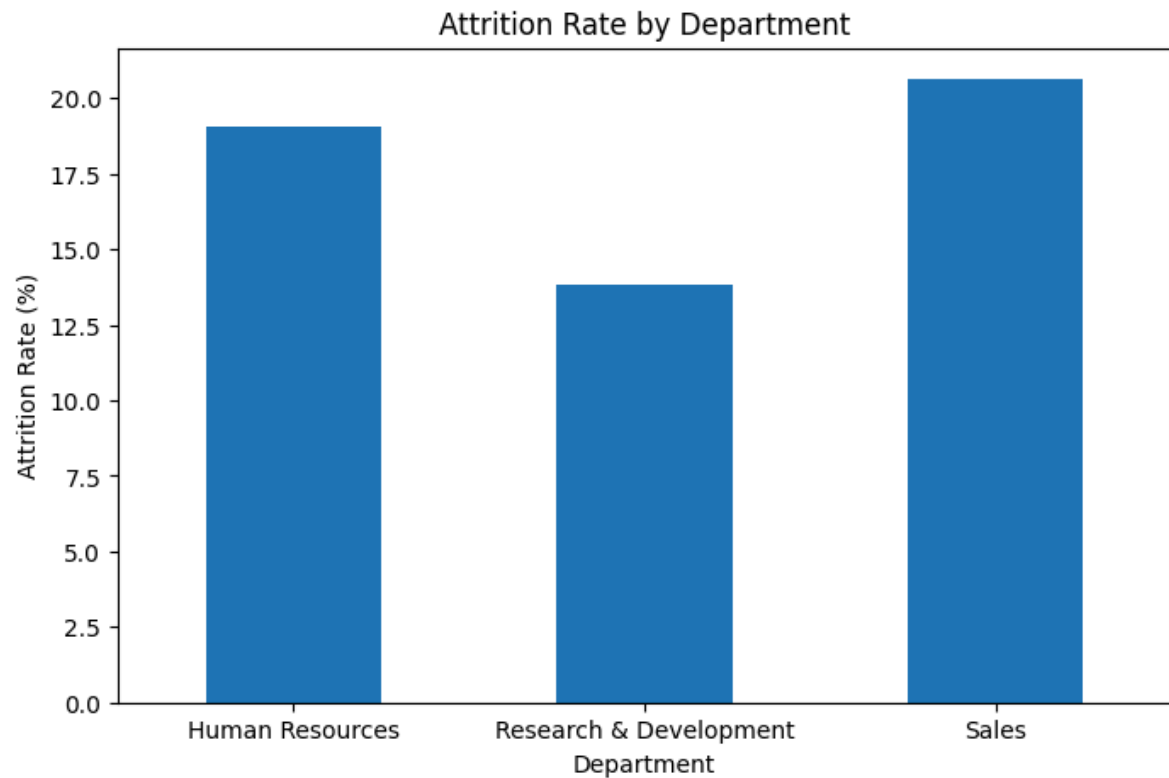
print(department_attrition.round(2))

department_attrition['Yes'].plot(
    kind='bar',
    figsize=(8,5),
    title='Attrition Rate by Department'
)

plt.xlabel('Department')
```

```
plt.ylabel('Attrition Rate (%)')
plt.xticks(rotation=0)
plt.show()
```

Attrition	No	Yes
Department		
Human Resources	80.95	19.05
Research & Development	86.16	13.84
Sales	79.37	20.63



13. Attrition by Job Role

Attrition rates were compared across different job roles to identify differences in employee turnover patterns.

```
In [107... role_attrition = pd.crosstab(
    df['JobRole'],
    df['Attrition'],
    normalize='index'
) * 100

print(role_attrition['Yes'].sort_values(ascending=False).round(2))

role_attrition['Yes'].sort_values().plot(
    kind='barh',
    figsize=(10,6),
    title='Attrition Rate by Job Role'
)

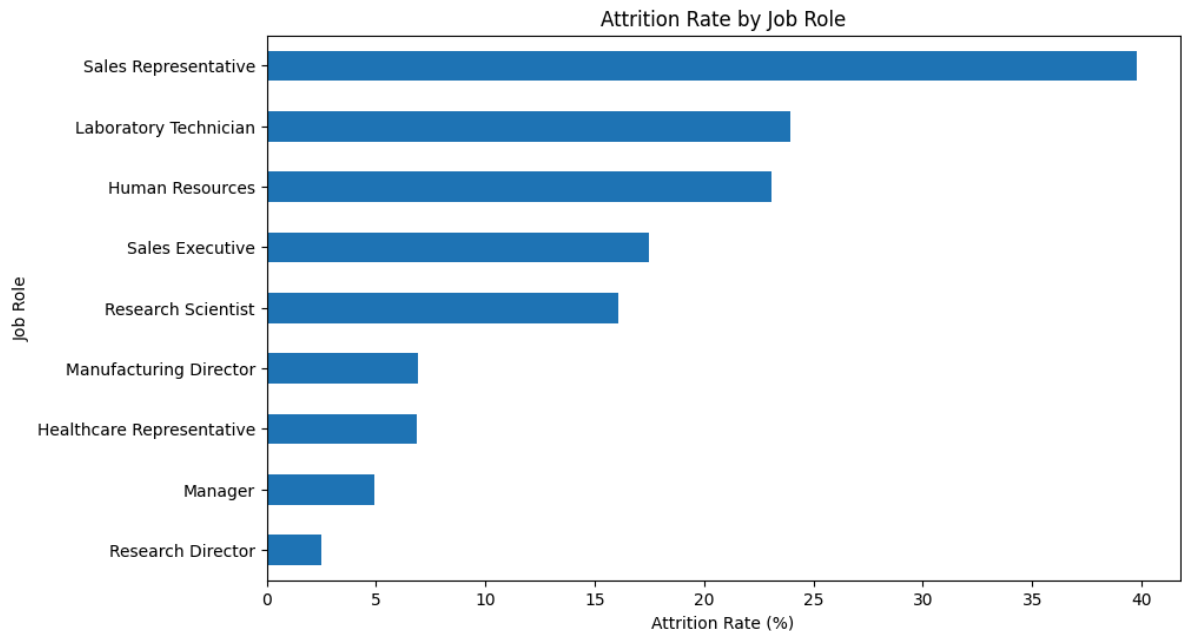
plt.xlabel('Attrition Rate (%)')
plt.ylabel('Job Role')
plt.show()
```

```

JobRole
Sales Representative      39.76
Laboratory Technician    23.94
Human Resources          23.08
Sales Executive          17.48
Research Scientist       16.10
Manufacturing Director   6.90
Healthcare Representative 6.87
Manager                  4.90
Research Director        2.50

```

Name: Yes, dtype: float64



14. Overtime and Attrition

The relationship between overtime status and employee attrition was examined by comparing attrition percentages for employees who work overtime and those who do not.

```

In [108... overtime_attrition = pd.crosstab(
    df['OverTime'],
    df['Attrition'],
    normalize='index'
) * 100

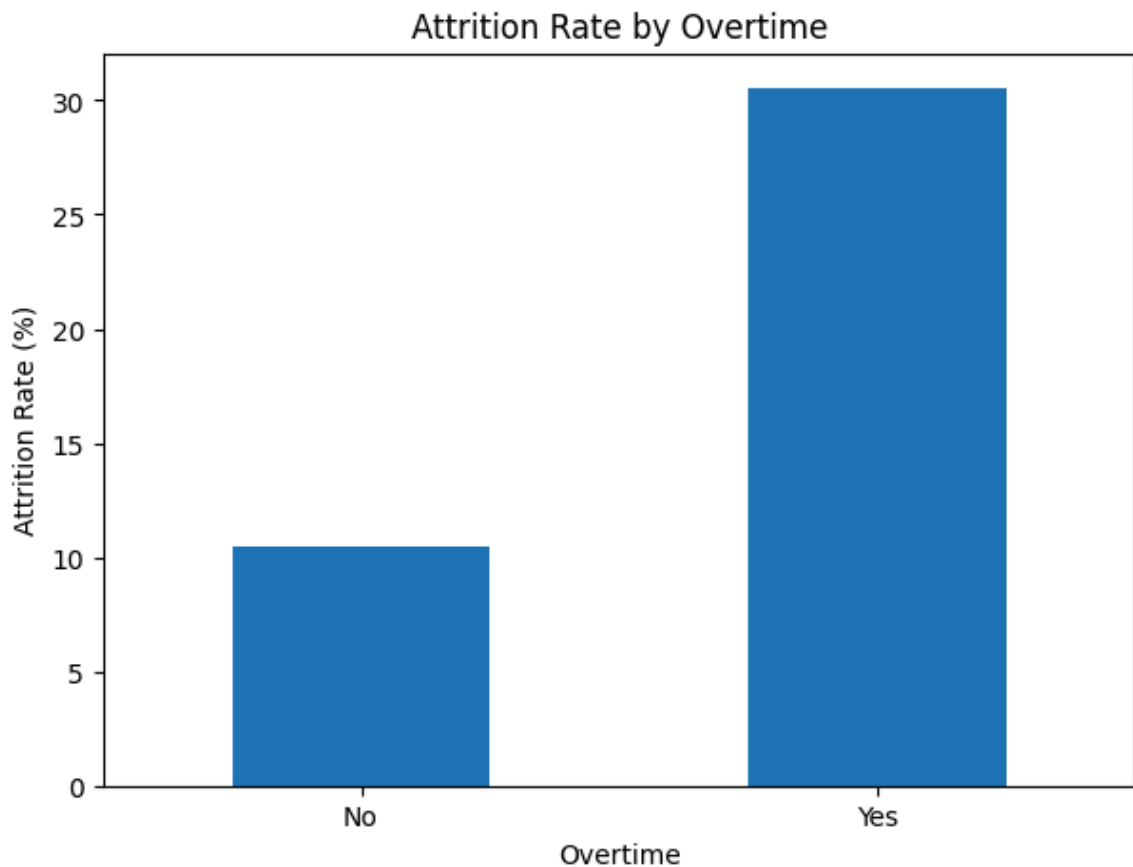
print(overtime_attrition.round(2))

overtime_attrition['Yes'].plot(
    kind='bar',
    figsize=(7,5),
    title='Attrition Rate by Overtime'
)

plt.xlabel('Overtime')
plt.ylabel('Attrition Rate (%)')
plt.xticks(rotation=0)
plt.show()

```

Attrition	No	Yes
OverTime		
No	89.56	10.44
Yes	69.47	30.53



15. Job Satisfaction and Attrition

Attrition was compared across the four job satisfaction levels to identify differences in employee attrition patterns.

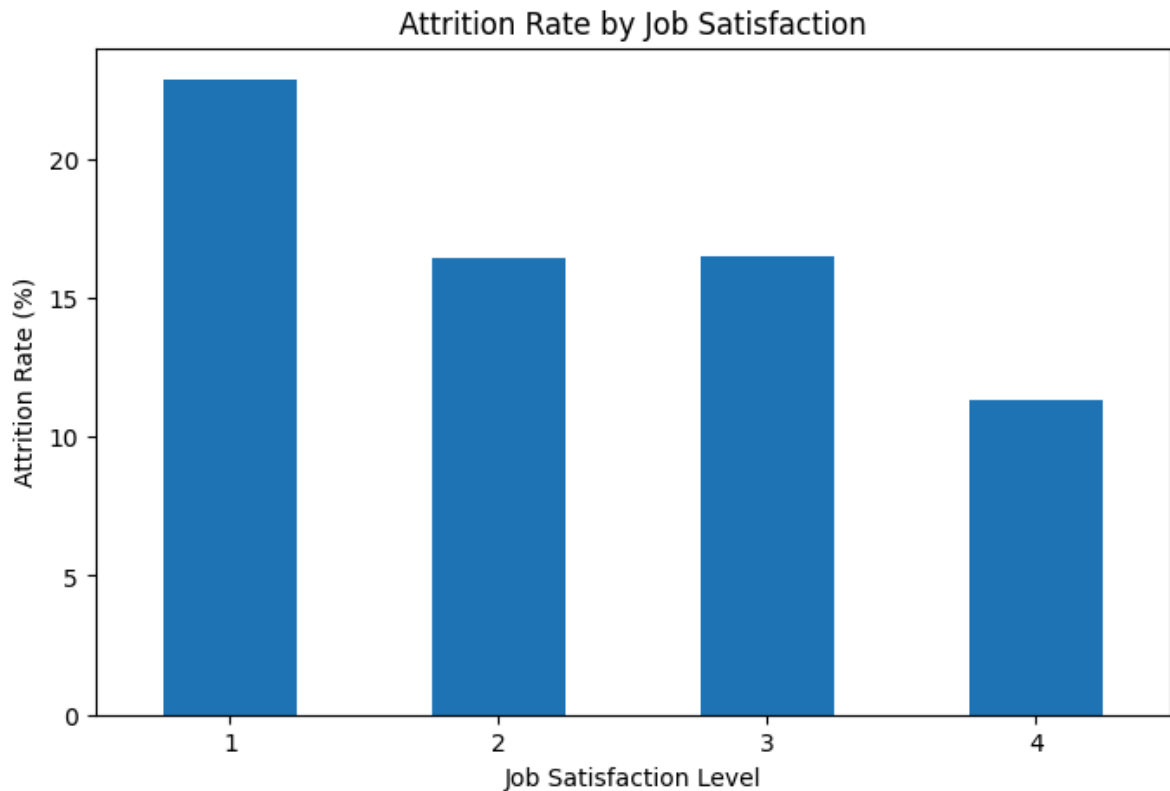
```
In [109... satisfaction_attrition = pd.crosstab(
    df['JobSatisfaction'],
    df['Attrition'],
    normalize='index'
) * 100

print(satisfaction_attrition.round(2))

satisfaction_attrition['Yes'].plot(
    kind='bar',
    figsize=(8,5),
    title='Attrition Rate by Job Satisfaction'
)

plt.xlabel('Job Satisfaction Level')
plt.ylabel('Attrition Rate (%)')
plt.xticks(rotation=0)
plt.show()
```

Attrition	No	Yes
JobSatisfaction		
1	77.16	22.84
2	83.57	16.43
3	83.48	16.52
4	88.67	11.33



16. Monthly Income and Attrition

Average monthly income was compared between employees who stayed and employees who left.

The comparison provides a descriptive view of income differences between the two attrition groups.

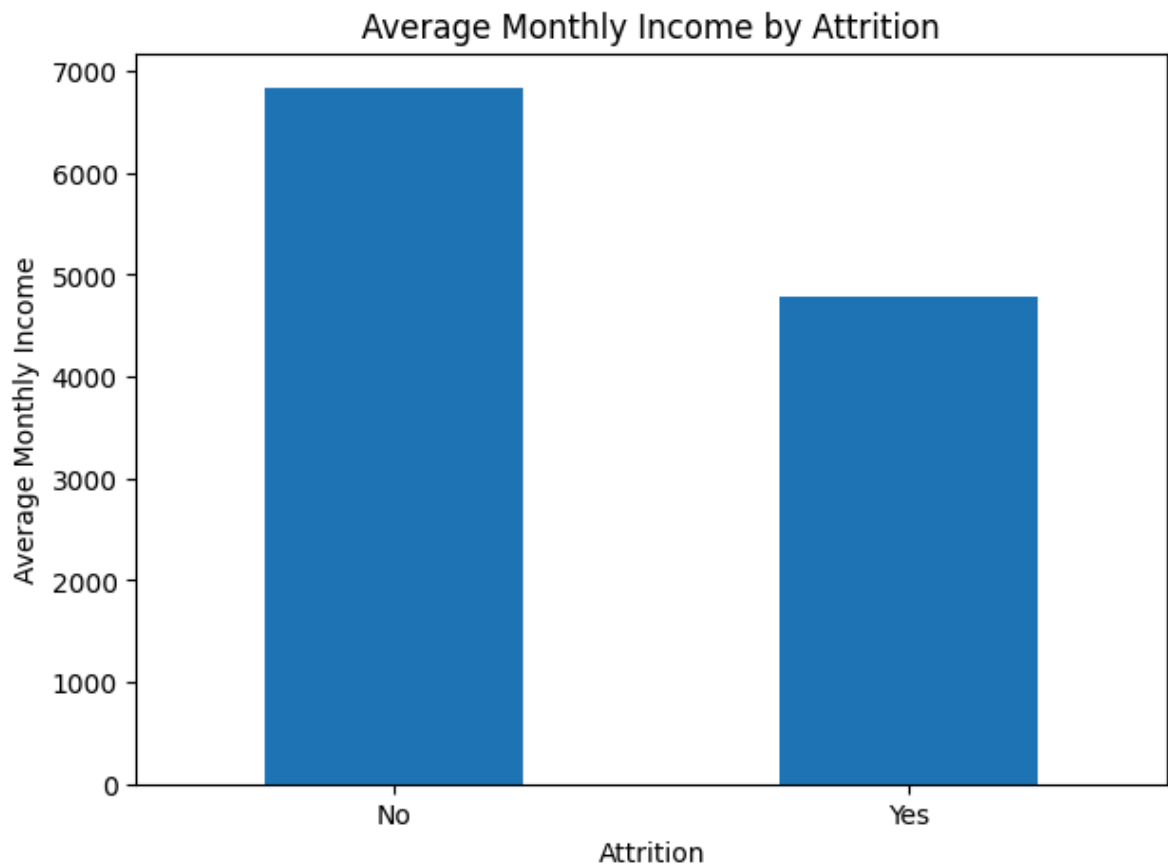
```
In [110... income_attrition = df.groupby('Attrition')['MonthlyIncome'].mean()

print(income_attrition.round(2))

income_attrition.plot(
    kind='bar',
    figsize=(7,5),
    title='Average Monthly Income by Attrition'
)

plt.xlabel('Attrition')
plt.ylabel('Average Monthly Income')
plt.xticks(rotation=0)
plt.show()
```

```
Attrition
No      6832.74
Yes     4787.09
Name: MonthlyIncome, dtype: float64
```



17. Years at Company and Attrition

Employee attrition was analyzed across different tenure groups to compare attrition patterns among employees with different years of experience in the organization.

```
In [111... bins = [0, 2, 5, 10, 20, 50]

labels = [
    '0-2',
    '3-5',
    '6-10',
    '11-20',
    '20+'
]

df['TenureGroup'] = pd.cut(
    df['YearsAtCompany'],
    bins=bins,
    labels=labels,
    include_lowest=True
)

tenure_attrition = pd.crosstab(
    df['TenureGroup'],
    df['Attrition'],
    normalize='index'
) * 100

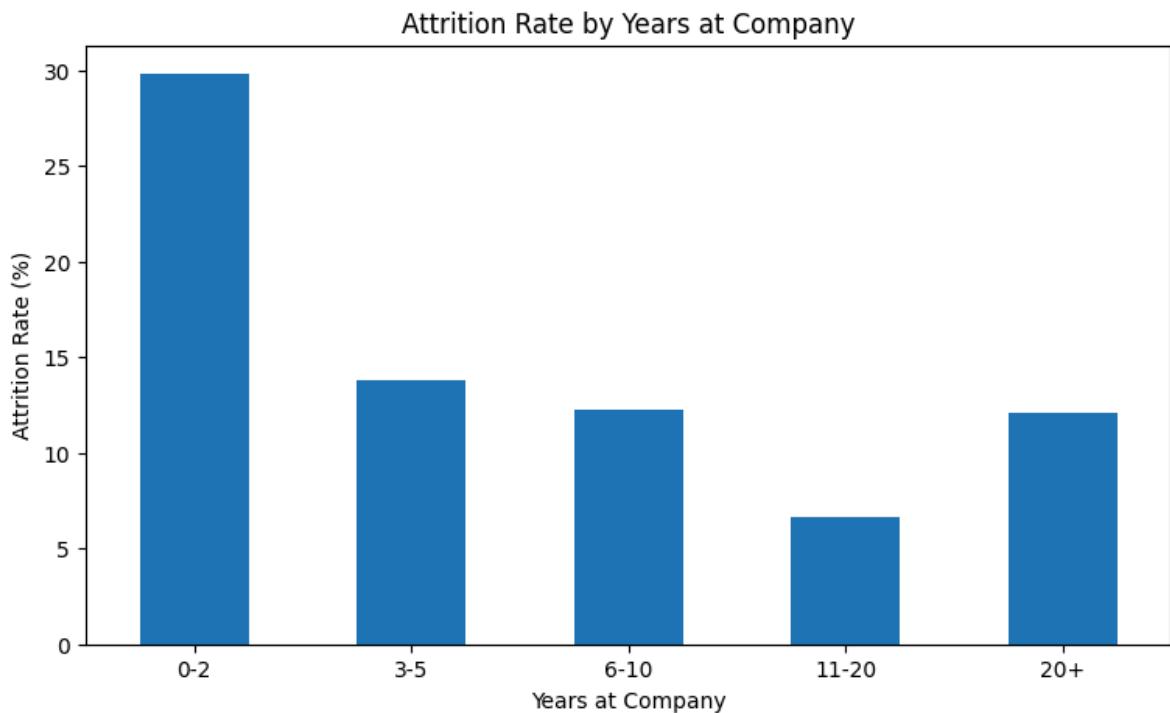
print(tenure_attrition.round(2))

tenure_attrition['Yes'].plot(
    kind='bar',
```

```
figsize=(9,5),
title='Attrition Rate by Years at Company'
)

plt.xlabel('Years at Company')
plt.ylabel('Attrition Rate (%)')
plt.xticks(rotation=0)
plt.show()
```

Attrition	No	Yes
TenureGroup		
0-2	70.18	29.82
3-5	86.18	13.82
6-10	87.72	12.28
11-20	93.33	6.67
20+	87.88	12.12



18. Years at Company vs Monthly Income

A scatter plot was used to examine the relationship between years at the company and monthly income while differentiating employees based on attrition status.

In [112...

```
plt.figure(figsize=(8,5))

sns.scatterplot(
    data=df,
    x='YearsAtCompany',
    y='MonthlyIncome',
    hue='Attrition',
    alpha=0.6
)

plt.title('Monthly Income vs Years at Company')
plt.xlabel('Years at Company')
plt.ylabel('Monthly Income')

plt.show()
```



19. Key Insights

- The dataset contains 1,470 employee records with no missing or duplicate records.
- Overall employee attrition is 16.12%.
- Sales has the highest department-level attrition at 20.63%.
- Sales Representatives show the highest job-role attrition at 39.76%.
- Employees working overtime show 30.53% attrition compared with 10.44% for employees without overtime.
- Job satisfaction level 1 has the highest attrition rate at 22.84%, while level 4 has 11.33%.
- Employees who left have a lower average monthly income of 4,787.09 compared with 6,832.74 for employees who stayed.
- Employees with 0–2 years at the company show the highest tenure-group attrition at 29.82%.

20. Recommendations

- Monitor attrition patterns in high-attrition job roles, particularly Sales Representatives.
- Review overtime requirements and workload distribution.
- Strengthen retention efforts for employees in their early years at the company.
- Monitor employee satisfaction levels and identify areas requiring improvement.
- Consider compensation, job role and tenure together when studying employee retention.

21. Conclusion

The Employee Attrition Analysis demonstrates how Exploratory Data Analysis can be used to understand workforce patterns.

The analysis covered data understanding, data cleaning, descriptive statistics, overall attrition, department and job-role comparisons, overtime, job satisfaction, monthly income and years at the company.

The findings provide a descriptive view of employee groups associated with different attrition patterns and can support further investigation into employee retention.

Overall, the project demonstrates the practical application of Python, Pandas and data visualization techniques to an HR analytics problem.

In [112...